Artificial Intelligence Degree at USJ

DATA ENGINEERING: CLOUD SERVICES

SUBJECT CURRICULUM

1 - Data Engineering Fundamentals
- 1.1 - What is Data Engineering
- 1.2 - Data Lifecycle
- 1.3 - Programming Languages
  - 1.3.1 - Python, R y Jupyter Notebooks
  - 1.3.2 - Other Languages for Data Engineering
- 1.4 - EL vs ETL vs ELT
- 1.5 - Data Governance

2 - Cloud Services
- 2.1 - Cloud vs On Premise
- 2.2 - Cloud Services Comparison
- 2.3 - Costs Engineering
- 2.4 - Specific Cloud Providers

3 - Distributed Systems
- 3.1 - Distributed Storage Systems
- 3.2 - Distributed Computing
- 3.3 - Event-Based architectures
- 3.4 - Orchestration
- 3.5 - Serverless Architectures

4 - Pre-Processing and Post-Processing
- 4.1 - Data access methods
- 4.2 - Raw Data and Modelling
- 4.3 - Data Cleaning
- 4.4 - Data Base Tunning

5 - Delivery
- 5.1 - Delivery methods
- 5.2 - Data Lakes and Data Mesh
- 5.3 - Data Marketplaces

6 - Deployment and Data Quality
- 6.1 - Fundamentals of Data Quality
- 6.2 - Monitoring
- 6.3 - CI/CD

7 - AI and Data Engineering
- 7.1 - Use of LLMs
- 7.2 - Data for Machine Learning

CLOUD VS ON PREMISE

SUBJECT CURRICULUM

1 - Data Engineering Fundamentals	2 - Cloud Services	3 - Distributed Systems
1.1 - What is Data Engineering	2.1 - Cloud vs On Premise	3.1 - Distributed Storage Systems
1.2 - Data Lifecycle	2.2 - Cloud Services Comparison	3.2 - Distributed Computing
1.3 - Programming Languages	2.3 - Costs Engineering	3.3 - Event-Based architectures
1.3.1 - Python, R y Jupyter Notebooks	2.4 - Specific Cloud Providers	3.4 - Orchestration
1.3.2 - Other Languages for Data Engineering		3.5 - Serverless Architectures
1.4 - EL vs ETL vs ELT		
1.5 - Data Governance		

ON PREMISES

Physical servers and storage systems owned and maintained by the organization

Complete control over hardware and software configurations

Data remains within the organization's physical boundaries

ON PREMISES

Full control over security and compliance

Predictable performance (no network latency to cloud)

No data egress costs

Suitable for highly sensitive data with strict regulatory requirements

ON PREMISES: CHALLENGES

- High upfront capital expenditure (CapEx)
- Limited scalability—adding capacity requires purchasing new hardware
- Maintenance overhead (hardware failures, upgrades, patches)
- Difficult to handle variable workloads efficiently
- Longer time to provision new resources

CLOUD

- Resources provided as services over the internet
- Pay-as-you-go pricing model
- Managed by cloud service providers

CLOUD

Elasticity: Scale compute and storage resources up or down based on demand

Cost Efficiency: Convert CapEx to OpEx; pay only for what you use

Speed: Provision resources in minutes instead of weeks/months

Global Reach: Deploy data pipelines closer to data sources/users worldwide

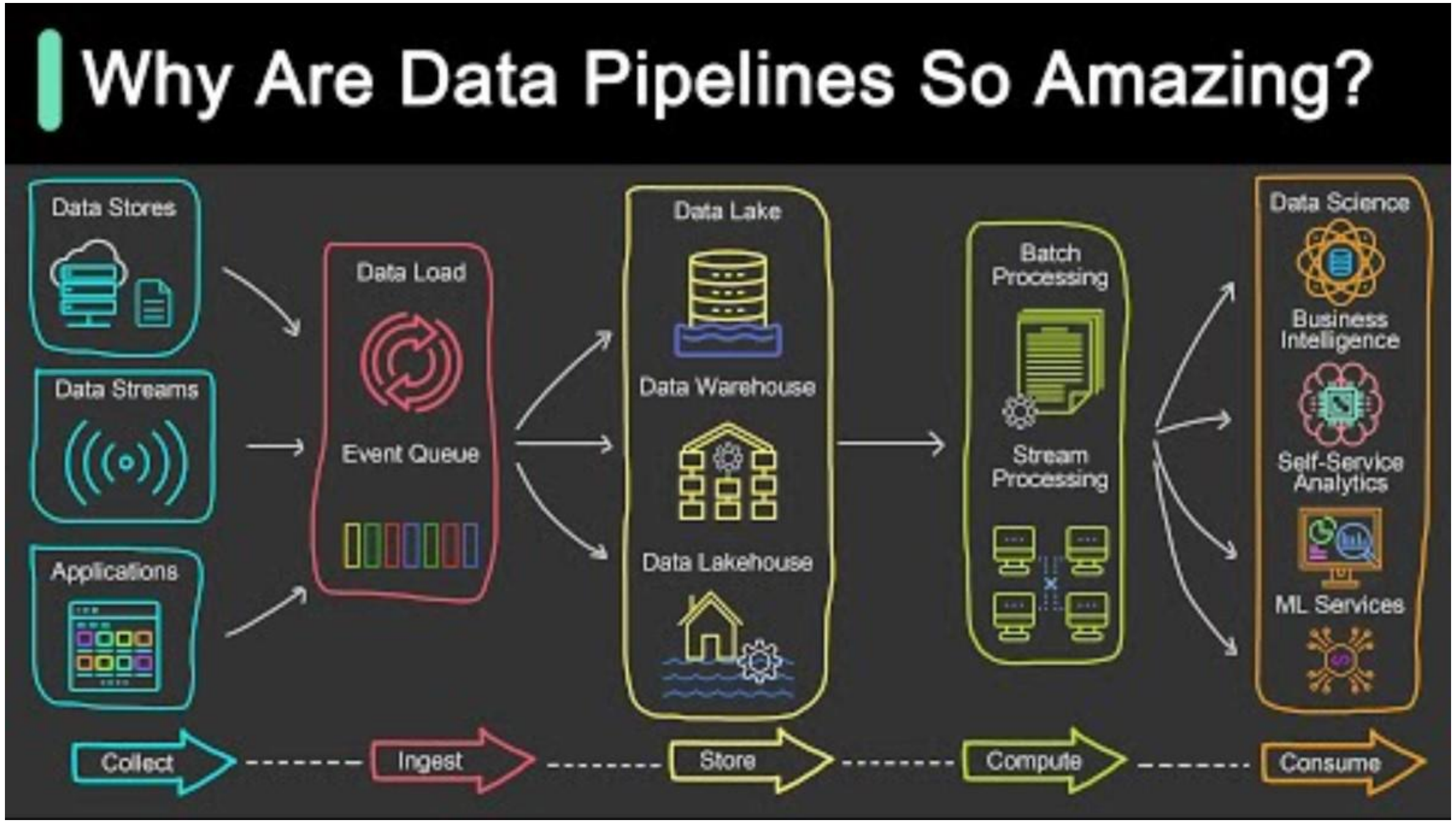
Managed Services: Focus on data engineering instead of infrastructure management

Innovation: Access to cutting-edge tools and services (ML/AI, streaming, etc.)

CLOUD: CHALLENGES

- Data transfer costs (especially egress)
- Potential vendor lock-in
- Learning curve for cloud-specific tools
- Security concerns (though often unfounded with proper configuration)
- Compliance considerations for certain industries

DATA PIPELINES REMINDER



KEY CONSIDERATIONS FOR DATA ENGINEERING

1. Data Gravity

- Where your data lives matter—moving large datasets between clouds is expensive and time-consuming
- Choose services close to your data sources

2. Cost Optimization

- Understand pricing models (compute, storage, data transfer)
- Use spot/preemptible instances for batch processing
- Implement data lifecycle policies

KEY CONSIDERATIONS FOR DATA ENGINEERING

3. Performance Requirements

- Consider latency requirements for real-time processing
- Evaluate managed services vs. self-managed for performance needs

4. Skills and Learning Curve

- Each cloud has its own tools and best practices
- Factor in team training and hiring considerations

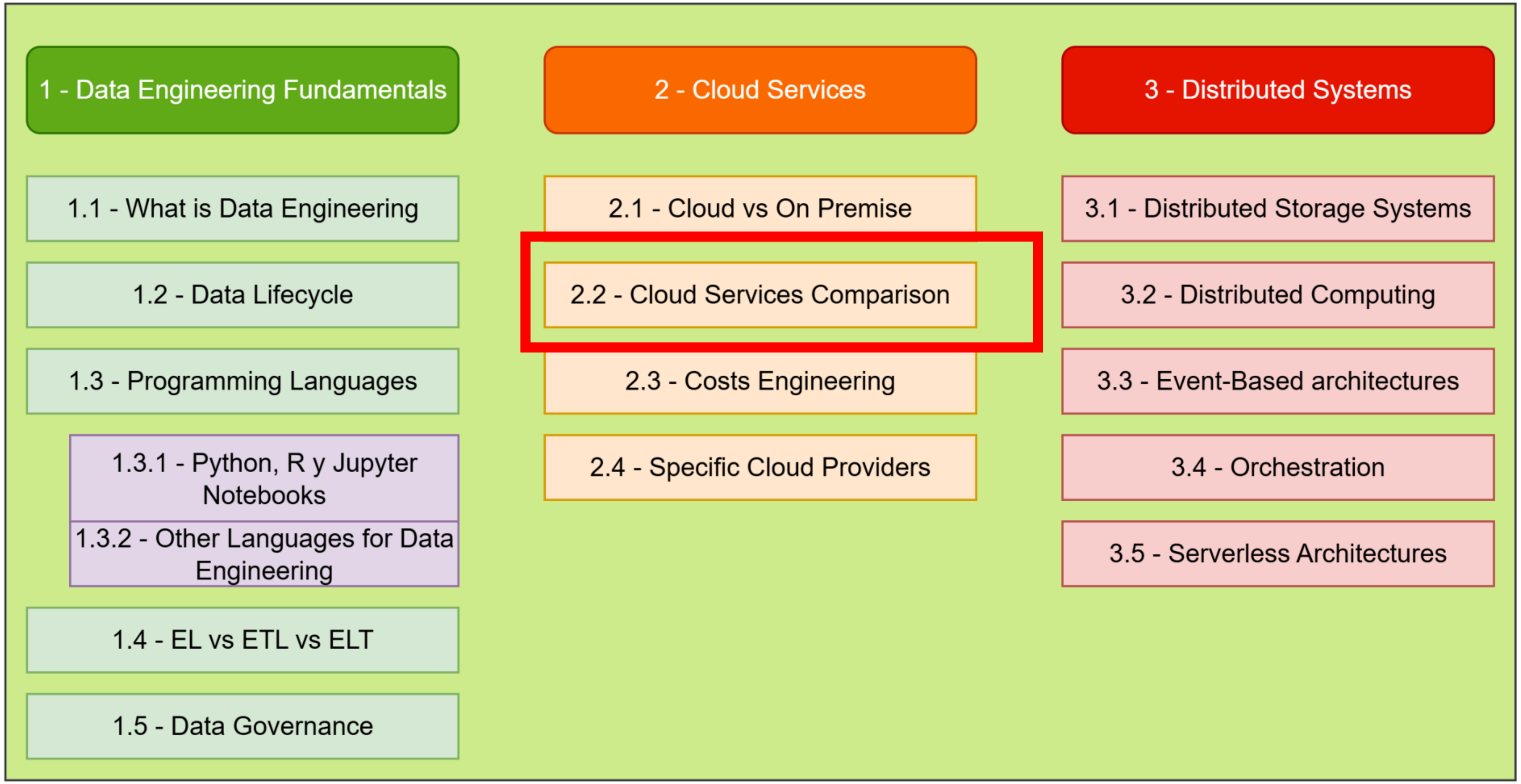
KEY CONSIDERATIONS FOR DATA ENGINEERING

5. Multi-Cloud Strategy

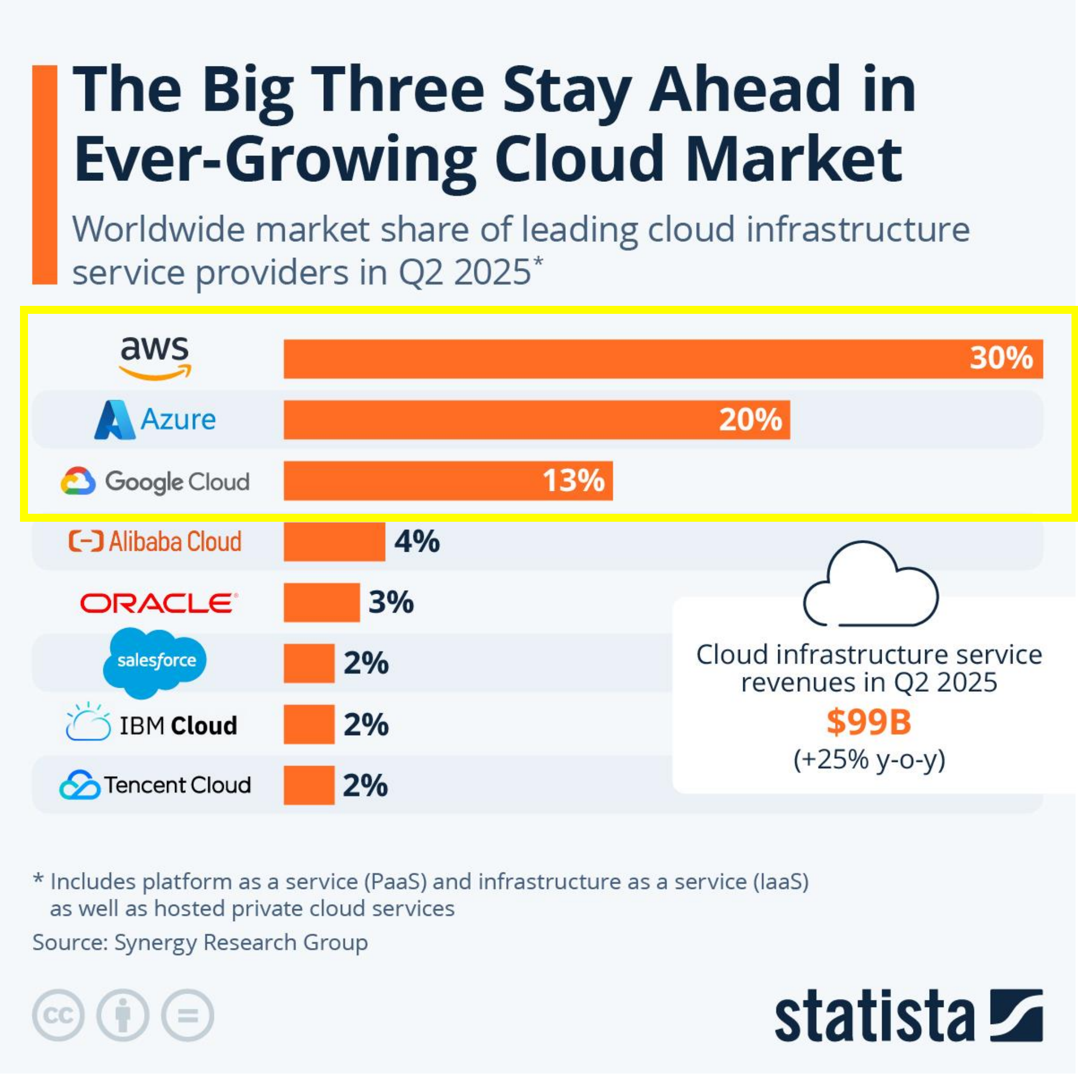
- Avoid vendor lock-in
- Use cloud-agnostic tools where possible (Kubernetes, Apache Spark)
- Consider data portability from day one

CLOUD SERVICES COMPARISON

SUBJECT CURRICULUM



MAJOR PLAYERS



MAJOR PLAYERS

Figure 1: Magic Quadrant for Strategic Cloud Platform Services

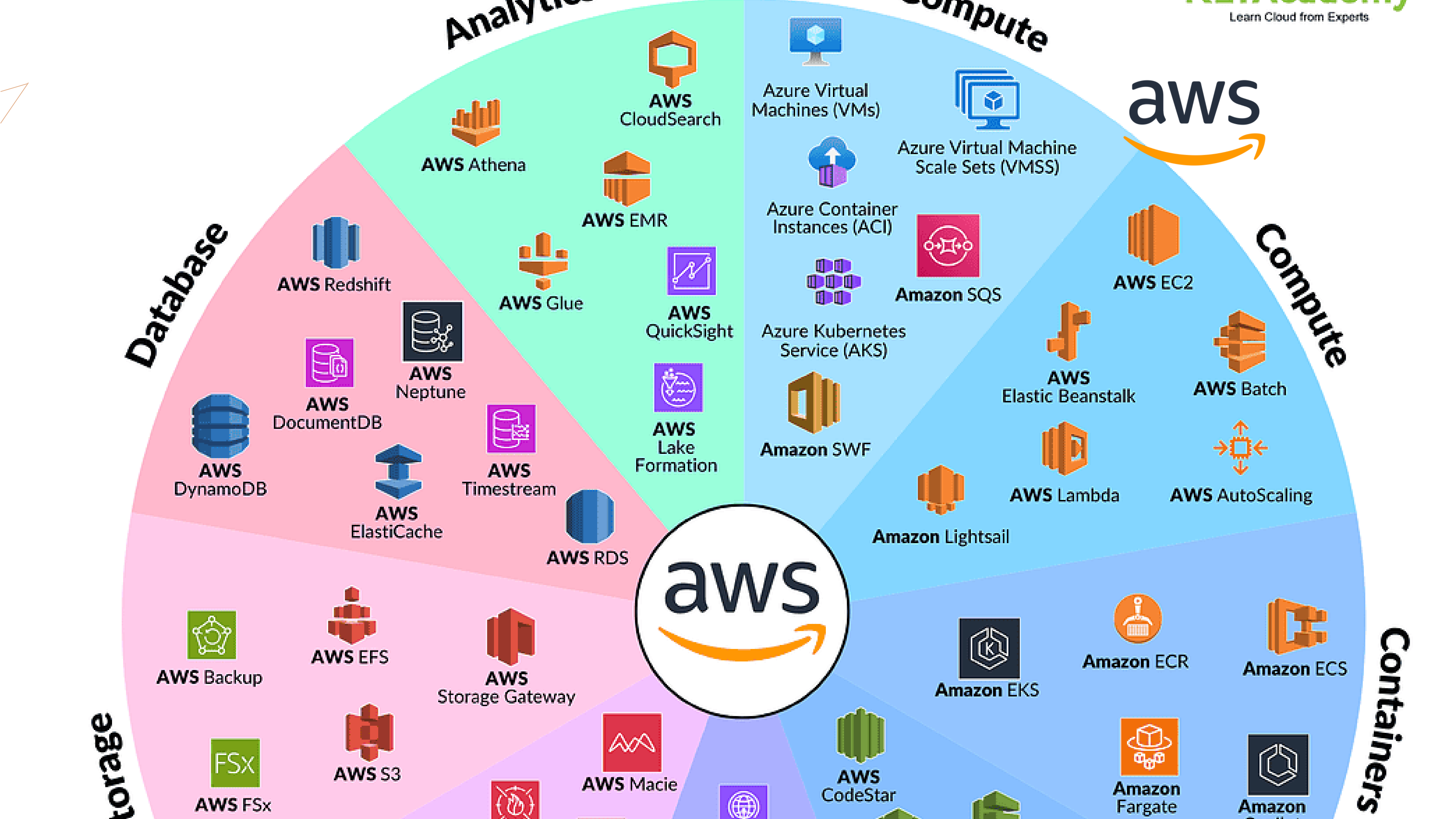
ENVIRONMENTAL IMPACT
ENVIRONMENTAL CONSIDERATIONS

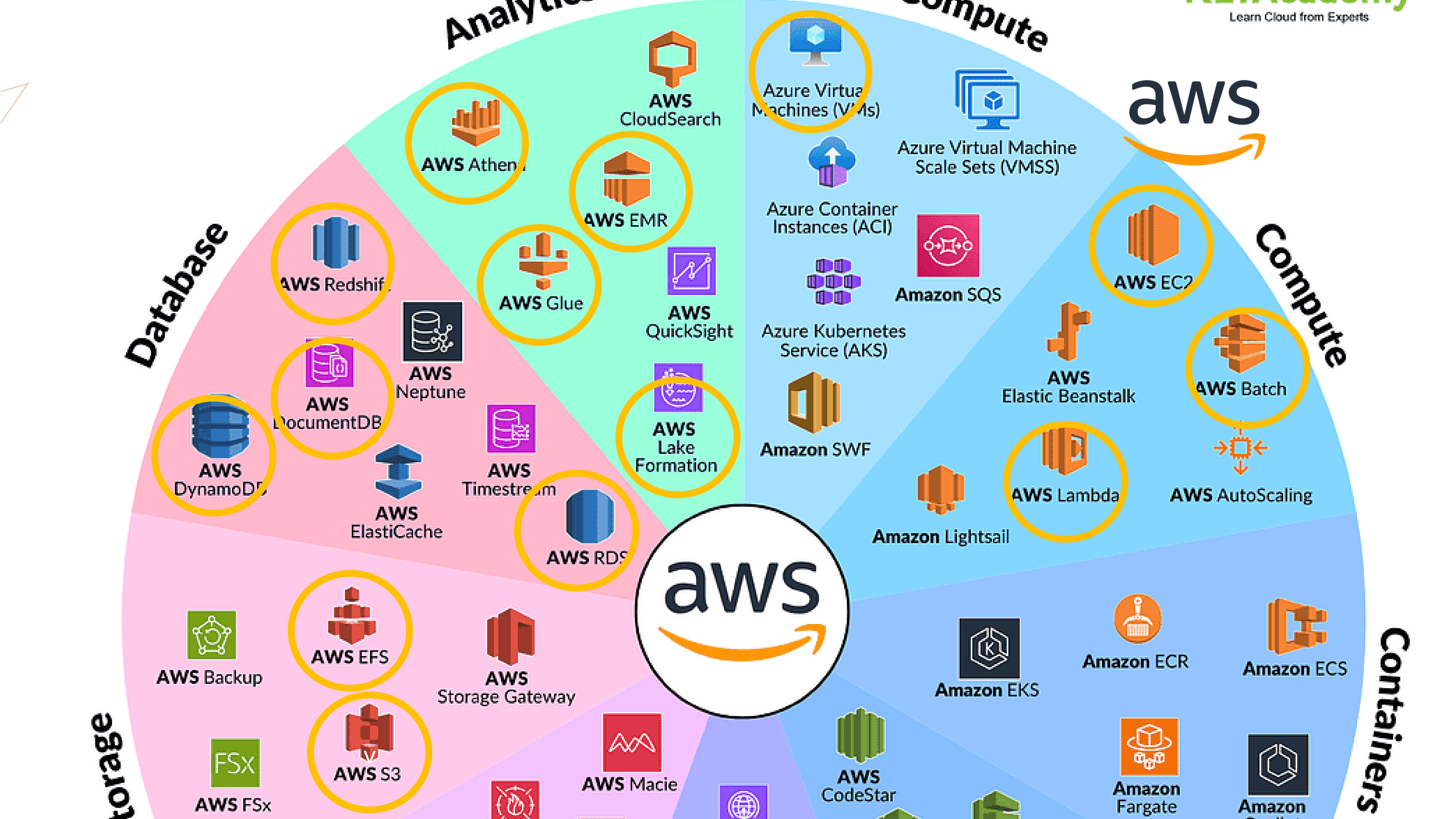
Energy Consumption & Carbon Footprint
•Data centers consume significant electricity for computing and cooling
•Geographic location affects carbon intensity (renewable vs. fossil fuel grids)
•Resource utilization efficiency directly impacts environmental cost

Provider Sustainability Commitments
•**AWS**: Net-zero carbon by 2040, 100% renewable energy by 2025
•**Microsoft Azure**: Carbon negative by 2030, zero waste by 2030
•**Google Cloud**: Carbon neutral since 2007, aiming for 24/7 renewable energy by 2030

K21Academy
Learn Cloud from Experts

aws


Learn Cloud from Experts
aws
Analytics
AWS CloudSearch
AWS Athena
AWS EMR
AWS Glue
AWS QuickSight
AWS Lake Formation
Compute
Azure Virtual Machines (VMs)
Azure Virtual Machine Scale Sets (VMSS)
Azure Container Instances (ACI)
Amazon SQS
Azure Kubernetes Service (AKS)
Amazon SWF
Compute
AWS EC2
AWS Elastic Beanstalk
AWS Batch
AWS Lambda
AWS AutoScaling
Amazon Lightsail
Database
AWS Redshift
AWS Neptune
AWS DocumentDB
AWS DynamoDB
AWS Timestream
AWS ElastiCache
AWS RDS
aws
Containers
Amazon EKS
Amazon ECR
Amazon ECS
AWS CodeStar
Amazon Fargate
Amazon
Storage
AWS Backup
AWS EFS
AWS Storage Gateway
AWS S3
AWS FSx
FSx
AWS Macie


Learn Cloud from Experts
aws
Analytics
Compute
Compute
Database
Containers
AWS CloudSearch
AWS Athena
AWS EMR
AWS Glue
AWS QuickSight
AWS Lake Formation
Azure Virtual Machines (VMs)
Azure Virtual Machine Scale Sets (VMSS)
Azure Container Instances (ACI)
Amazon SQS
Azure Kubernetes Service (AKS)
Amazon SWF
Amazon Lightsail
AWS EC2
AWS Elastic Beanstalk
AWS Batch
AWS Lambda
AWS AutoScaling
AWS Redshift
AWS Neptune
AWS DocumentDB
AWS DynamoDB
AWS ElastiCache
AWS Timestream
AWS RDS
aws
AWS Backup
AWS EFS
AWS Storage Gateway
AWS FSx
AWS S3
AWS Macie
AWS CodeStar
Amazon EKS
Amazon ECR
Amazon ECS
Amazon Fargate
Amazon

KEY AWS SERVICES FOR DATA ENGINEERING

Amazon S3

AWS Glue

AWS Redshift

Amazon EMR

Amazon Athena

Amazon Kinesis

AWS Lambda

Amazon QuickSight

PRACTICE AWS

To practice with some AWS services, you can register a free account in https://skillbuilder.aws/learn There you can find labs and courses that allows you to interact with the AWS console and the different services.

Recommended course: Introduction to AWS lambda
Lambda is the equivalent to Cloud Run Functions in GCP and Azure Functions in Azure. It allows you to execute code 'serverless': the code is uploaded to azure but only used in a machine when requested (Via HTTP, or some cloud event). Is not stored in any server unless it is called.

Open data sets in AWS

AWS Pricing Calculator

GOOGLE CLOUD (GCP)

Google Cloud

Compute

- Compute Engine
- App Engine
- Container Engine
- Container Registry
- Cloud Functions

Identity & Security

- Cloud IAM
- Cloud Resource Manager
- Cloud Security Scanner
- Cloud Platform Security

Networking

- Cloud Virtual Network
- Cloud Load Balancing
- Cloud CDN
- Cloud Interconnect
- Cloud DNS

Big Data

- BigQuery
- Cloud Dataflow
- Cloud Dataproc
- Cloud Datalab
- Cloud Pub/Sub
- Genomics

Storage and Databases

- Cloud Storage
- Cloud Bigtable
- Cloud Datastore
- Cloud SQL
- Persistent Disk

Machine Learning

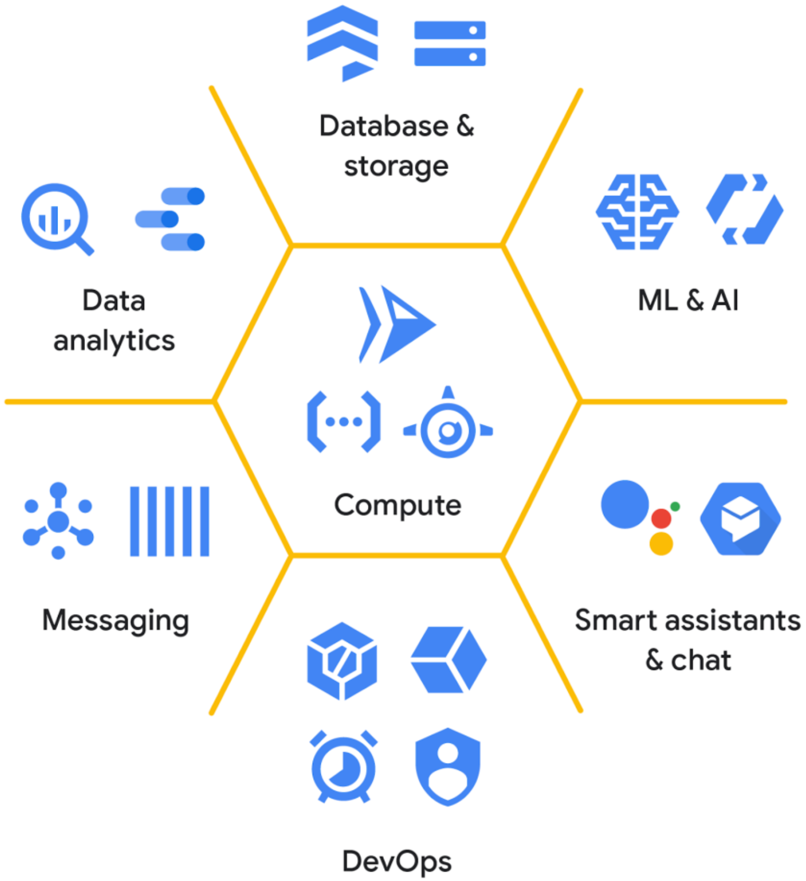
- Cloud Machine Learning
- Vision API
- Speech API
- Natural Language API
- Translation API
- Jobs API

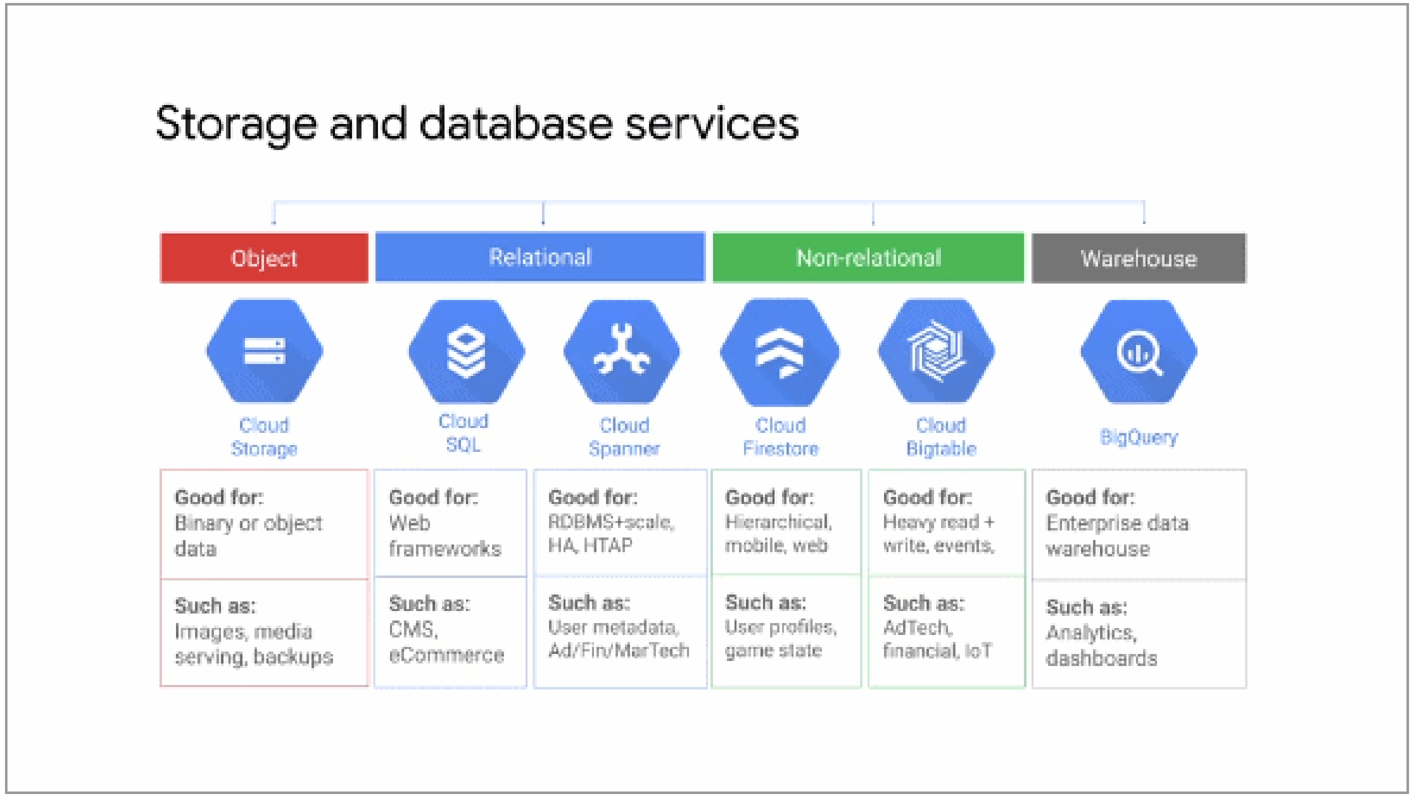
- Data engineering
- Useful/must

GOOGLE CLOUD (GCP)

Google Cloud

GOOGLE CLOUD (GCP)





PRACTICE GCP

Google Cloud

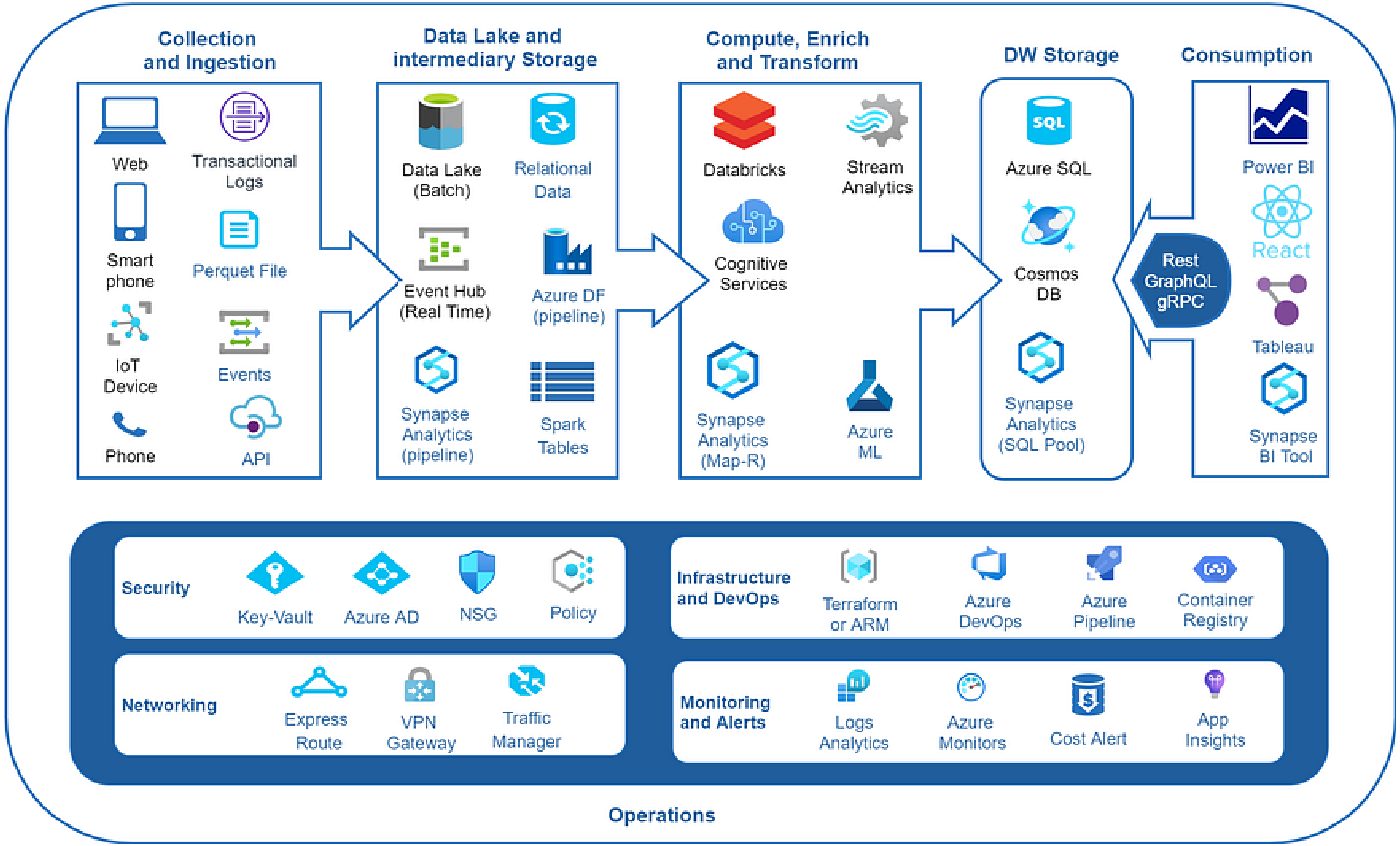
To practice with some GCP services, you can register a free account in https://www.cloudskillsboost.google/
There you can find labs and courses that allows you to interact with the GCP console and the different services.

Recommended course:
https://www.cloudskillsboost.google/focuses/12374?catalog_rank=%7B%22rank%22%3A13%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&parent=catalog&search_id=56291646

Open data sets in GCP

Google Cloud Pricing Calculator

AZURE



Collection and Ingestion
Web
Transactional Logs
Smart phone
Perquet File
IoT Device
Events
Phone
API
Data Lake and intermediary Storage
Data Lake (Batch)
Relational Data
Event Hub (Real Time)
Azure DF (pipeline)
Synapse Analytics (pipeline)
Spark Tables
Compute, Enrich and Transform
Databricks
Stream Analytics
Cognitive Services
Synapse Analytics (Map-R)
Azure ML
DW Storage
SQL
Azure SQL
Cosmos DB
Synapse Analytics (SQL Pool)
Consumption
Power BI
React
Tableau
Synapse BI Tool
Rest GraphQL gRPC
Security
Key-Vault
Azure AD
NSG
Policy
Infrastructure and DevOps
Terraform or ARM
Azure DevOps
Azure Pipeline
Container Registry
Networking
Express Route
VPN Gateway
Traffic Manager
Monitoring and Alerts
Logs Analytics
Azure Monitors
Cost Alert
App Insights
Operations

WORK WITH AZURE

To work with azure, we will need to get a student subscription: https://azure.microsoft.com/es-es/free/students , that has 100$ of free usage, fair enough for this course.
Also we will use Microsoft learn to access free courses: https://learn.microsoft.com/en-us/training/

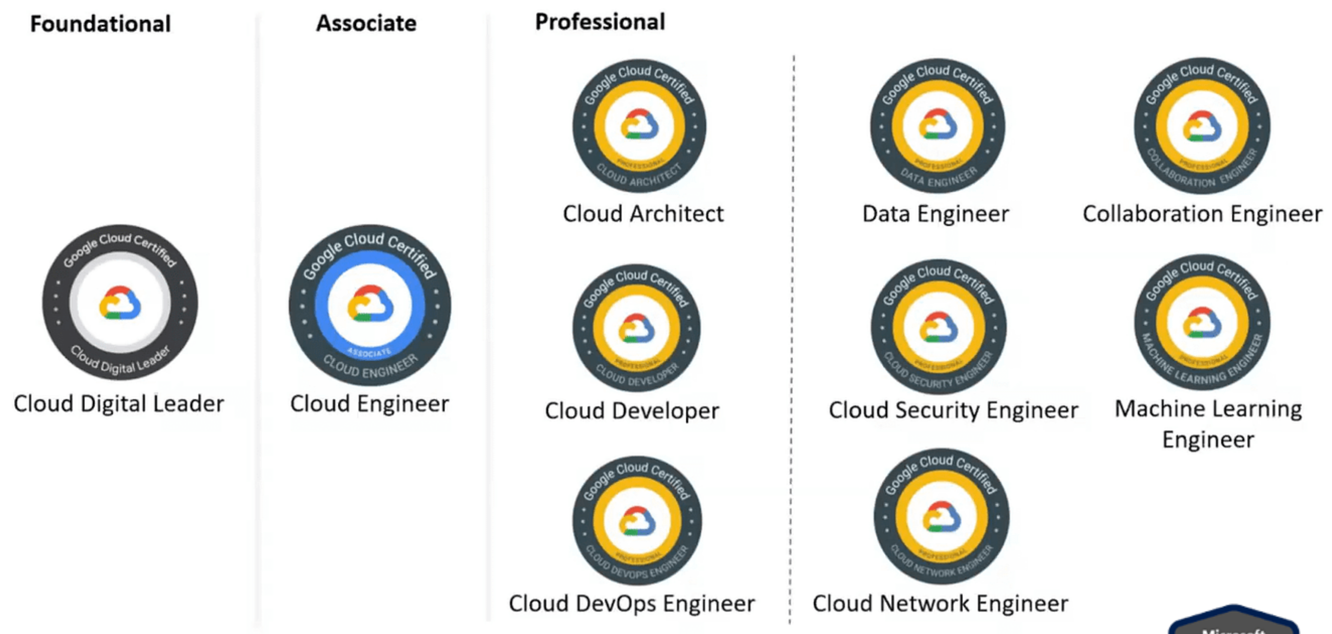
Recommended course: Azure DataBricks Solution

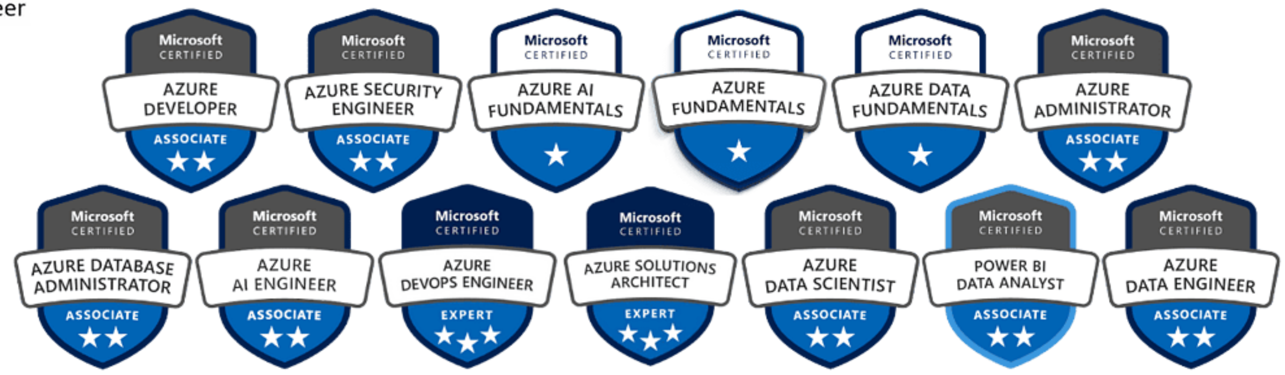
Databricks is a processing service, developed on top of Apache Spark, ready to work with large amounts of data. In this course we will work wth incremental data processing using Delta Live Tables (DLT) in Azure Databricks. It offers a robust and efficient way to manage and process large volumes of data by only processing the changes (deltas) since the last update.

Pricing Calculator | Microsoft Azure

‘EQUIVALENT’ SERVICES

CERTIFICATIONS





CERTIFICATIONS

Foundational

Cloud Digital Leader

Associate

Cloud Engineer

Professional

Cloud Architect

Data Engineer

Collaboration Engineer

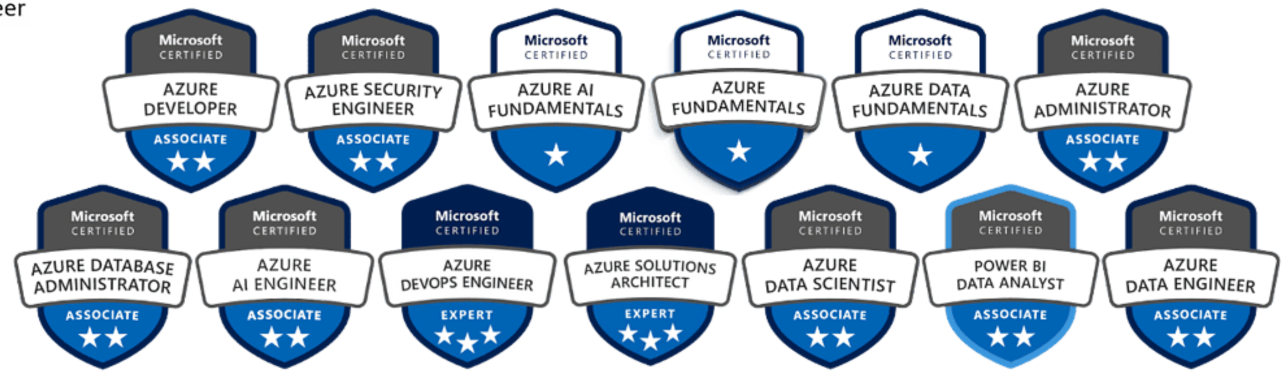
Cloud Developer

Cloud Security Engineer

Machine Learning Engineer

Cloud DevOps Engineer

Cloud Network Engineer



MAIN CLOUD SERVICES COMPARISONS:
OBJECT STORAGE

- **S3** - Highly durable object storage with multiple storage classes
- **S3 Glacier** - Long-term archival storage
- **EFS** - Managed elastic file storage

- **Blob Storage** - Massively scalable object storage for unstructured data
- **Data Lake Storage Gen2** - Hierarchical storage for analytics
- **Azure Files** - Managed file shares

- **Cloud Storage** - Unified object storage with automatic tiering
- **Filestore** - Managed network file storage
- **Persistent Disk** - Block storage for VMs

MAIN CLOUD SERVICES COMPARISONS:
RELATIONAL DATABASES

- **RDS** - Managed relational databases (MySQL, PostgreSQL, SQL Server , Oracle, MariaDB)
- **Aurora** - MySQL and PostgreSQL-compatible high-performance database
- **RDS Proxy** - Connection pooling for databases

- **Azure SQL Database** - Fully managed SQL database engine
- **Azure Database for PostgreSQL** - Managed PostgreSQL service
- **Azure Database for MySQL** - Managed MySQL service

- **Cloud SQL** - Managed MySQL, PostgreSQL, and SQL Server
- **Cloud Spanner** - Globally distributed relational database
- **AlloyDB** - PostgreSQL-compatible high-performance database

MAIN CLOUD SERVICES COMPARISONS:
BATCH PROCESSING

- **AWS Batch** - Fully managed batch processing at any scale
- **EMR** - Managed Hadoop/Spark for big data processing
- **Glue** - Serverless ETL service for batch jobs

- **Azure Batch** - Cloud-scale job scheduling and compute management
- **HDInsight** - Managed Hadoop, Spark, Hive cluster service
- **Azure Databricks** - Apache Spark-based analytics platform

- **Dataproc** - Managed Spark and Hadoop service for batch processing
- **Dataflow** - Batch data processing (Apache Beam)
- **Dataprep** - Visual data preparation tool

MAIN CLOUD SERVICES COMPARISONS:
STREAMING

- **Kinesis Data Streams** - Real-time data streaming platform
- **Kinesis Data Analytics** - Real-time analytics using SQL or Apache Flink
- **MSK** (Managed Streaming for Kafka) - Managed Apache Kafka service

- **Stream Analytics** - Real-time analytics service for streaming data
- **Event Hubs** - Big data streaming platform and event ingestion
- **HDInsight with Kafka** - Managed Kafka clusters

- **Dataflow** - Unified stream and batch data processing (Apache Beam)
- **Pub/Sub** - Real-time messaging and event streaming
- **Datastream** - Serverless change data capture (CDC) service

MAIN CLOUD SERVICES COMPARISONS:
DATA WAREHOUSE

- **Redshift** - Petabyte-scale data warehouse with columnar storage
- **Redshift Spectrum** - Query S3 data directly from Redshift
- **Athena** - Serverless interactive query service for S3

- **Synapse Analytics** - Analytics service combining data warehouse and big data
- **Synapse Serverless SQL Pool** - Query data without provisioning
- **Azure Analysis Services** - Enterprise-grade analytics engine

- **BigQuery** - Serverless, highly scalable data warehouse with SQL interface
- **BigLake** - Unified storage engine across data lakes and warehouses
- **Looker** - Business intelligence and analytics platform

MAIN CLOUD SERVICES COMPARISONS:
MONITORING

- **CloudWatch** - Monitoring and observability for resources and applications
- **X-Ray** - Distributed tracing for application analysis
- **CloudTrail** - API activity logging and governance

- **Azure Monitor** - Full-stack monitoring service
- **Application Insights** - Application performance management (APM)
- **Log Analytics** - Log collection and analysis

- **Cloud Monitoring** - Infrastructure and application monitoring
- **Cloud Logging** - Log management and analysis
- **Cloud Trace** - Distributed tracing system

MAIN CLOUD SERVICES COMPARISONS:
ORCHESTRATION

- **Step Functions** - Serverless workflow orchestration using visual workflows
- **Managed Workflows for Apache Airflow (MWAA)** - Managed Airflow service
- **AWS Glue Workflows** - ETL workflow orchestration

- **Data Factory** - Cloud ETL/ELT service with visual data pipeline creation
- **Logic Apps** - Workflow automation and integration platform
- **Azure Databricks Workflows** - Job orchestration within Databricks

- **Cloud Composer** - Managed Apache Airflow for workflow orchestration
- **Cloud Scheduler** - Cron job scheduling service
- **Workflows** - Serverless workflow orchestration

MAIN CLOUD SERVICES COMPARISONS:
SECURITY

- **IAM** - Identity and access management
- **Security Hub** - Unified security and compliance dashboard
- **GuardDuty** - Threat detection service
- **Secrets Manager** - Secrets and credentials management

- **Azure AD (Entra ID)** - Identity and access management
- **Security Center** (Defender for Cloud) - Security posture management
- **Sentinel** - Cloud-native SIEM and SOAR
- **Key Vault** - Secrets and key management

- **Cloud IAM** - Access control and identity management
- **Security Command Center** - Security and risk management platform
- **Secret Manager** - Store and manage secrets
- **Chronicle** - Security analytics platform

MAIN CLOUD SERVICES COMPARISONS:
GOVERNANCE

- **Lake Formation** - Data lake governance and access control
- **Glue Data Catalog** - Metadata repository and data catalog
- **Macie** - Data security and privacy service
- **Config** - Resource configuration tracking

- **Purview** - Unified data governance for on-premises and cloud data
- **Policy** - Enforce organizational standards and compliance
- **Blueprints** - Repeatable environment definitions

- **Dataplex** - Intelligent data fabric for unified governance
- **Data Catalog** - Metadata management and data discovery
- **Policy Intelligence** - Policy management and recommendations

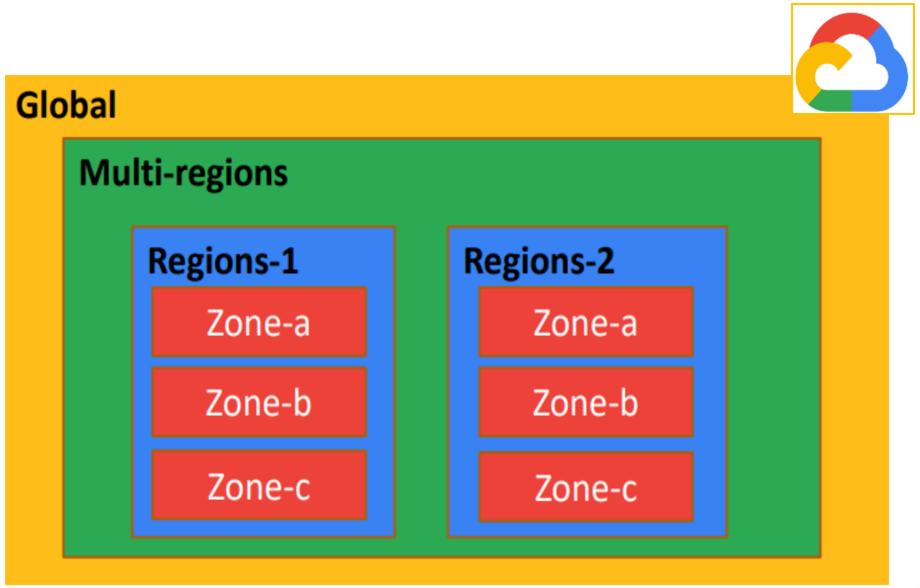
MAIN CLOUD SERVICES COMPARISONS:
MACHINE LEARNING

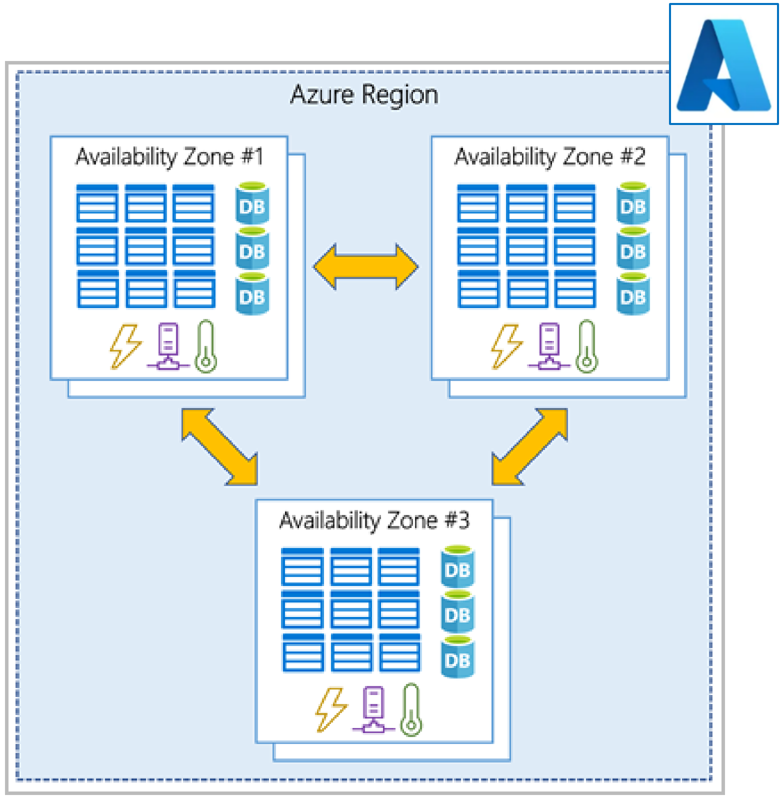
- **SageMaker** - Complete ML platform for building, training, and deploying models
- **Bedrock** - Managed service for foundation models
- **Rekognition** - Image and video analysis
- **Comprehend** - Natural language processing service

- **Azure Machine Learning** - End-to-end ML service with AutoML and MLOps
- **Cognitive Services** - Pre-built AI APIs (vision, speech, language)
- **Azure OpenAI Service** - Access to OpenAI models
- **Azure Databricks ML** - ML on Databricks platform

- **Vertex AI** - Unified ML platform with pre-trained models and custom training
- **AutoML** - Automated machine learning model training
- **AI Platform Notebooks** - Managed JupyterLab environment
- **Pre-trained APIs** - Vision, Natural Language, Translation, Speech APIs

MAIN CLOUD SERVICES COMPARISONS:
REGIONS





MAIN CLOUD SERVICES COMPARISONS: REGIONS

Dimension	AWS	Azure	GCP
Number of Regions	33+ regions	60+ regions	40+ regions
Availability Zones	3-6 AZs per region, physically separate data centers	3+ AZs per region (where available), some regions have single zone	3+ zones per region, similar to AWS model
Edge Locations	400+ CloudFront edge locations	170+ CDN PoPs	140+ edge locations
Global Coverage	Strongest in Americas and Asia	Best global coverage overall, especially emerging markets	Strong but fewer regions than competitors
Naming Convention	us-east-1, eu-west-2	East US, West Europe	us-central1, europe-west1

MAIN CLOUD SERVICES COMPARISONS:
IDENTITY AND ACCESS MANAGEMENT (IAM)

Dimension	AWS	Azure	GCP
IAM Model	IAM users, groups, roles with policy documents	Azure AD (Entra ID) with RBAC	Cloud IAM with roles and service accounts
Policy Language	JSON-based policy documents	Azure Policy (JSON) + RBAC	IAM Policy binding, condition expressions
Federation	SAML 2.0, OIDC support	Native AD federation, strong enterprise identity	OIDC, SAML, Workload Identity Federation
Service Identity	IAM roles for services	Managed identities	Service accounts
Granularity	Very granular, can be complex	Good balance of granularity and simplicity	Fine-grained, simpler than AWS
Root Account	Root user (discouraged)	Global Administrator	Organization Owner
MFA Support	Yes, multiple options	Yes, strong MFA including passwordless	Yes, including hardware tokens

MAIN CLOUD SERVICES COMPARISONS:
PRICING AND BILLING

Dimension	AWS	Azure	GCP
Billing Granularity	Per second (most services)	Per minute to per hour depending on service	Per second for most services
Reserved Instances	1 or 3 years commitments, up to 72% savings	Reserved VMs, 1-3 years, up to 72% savings	Committed use discounts, 1-3 years, up to 57% savings
Spot/Preemptible	Spot instances (up to 90% off)	Spot VMs (up to 90% off)	Preemptible VMs (up to 80% off)
Sustained Use	No automatic discounts	No automatic discounts	Automatic sustained use discounts (up to 30%)
Free Tier	12 months + always free tier	12 months + some always free	90-day $300 credit + always free
Cost Tools	Cost Explorer, Budgets, Trusted Advisor	Cost Management + Billing, Advisor	Cost Management, Recommender
Egress Costs	Expensive	Expensive	Expensive (slightly better for some services)
Pricing Transparency	Complex, many SKUs	Complex, calculator helps	Generally simpler, more transparent

MAIN CLOUD SERVICES COMPARISONS:
COMPUTE SERVICES

Dimension	AWS	Azure	GCP
VMs	EC2 - 500+ instance types	Virtual Machines - 700+ sizes	Compute Engine - 60+ machine types
VM Customization	Limited (nitro instances)	Limited custom sizes	Custom machine types (flexible vCPU/memory)
Containers	ECS, EKS, Fargate	AKS, Container Instances, App Service	GKE (best Kubernetes experience), Cloud Run
Serverless	Lambda (most mature)	Azure Functions	Cloud Functions, Cloud Run
Bare Metal	Limited (i3.metal)	Azure Dedicated Host	Bare Metal Solution (limited)
Auto-scaling	EC2 Auto Scaling, comprehensive	VM Scale Sets, comprehensive	Managed Instance Groups, good
Windows Support	Excellent	Best (native Windows integration)	Good but not primary focus

MAIN CLOUD SERVICES COMPARISONS: NETWORKING

Dimension	AWS	Azure	GCP
Virtual Network	VPC (per region)	VNet (per region)	VPC (global resource)
Subnets	Per AZ	Per address space	Per region
Network Philosophy	Regional isolation	Hub-spoke common	Global by default
Private Connectivity	Direct Connect	ExpressRoute	Cloud Interconnect
VPN	Site-to-Site VPN, Client VPN	VPN Gateway	Cloud VPN
Load Balancing	ELB (ALB, NLB, CLB, GWLB)	Azure Load Balancer, Application Gateway	Cloud Load Balancing (global, single anycast IP)
DNS	Route 53	Azure DNS, Traffic Manager	Cloud DNS
CDN	CloudFront (mature)	Azure CDN, Front Door	Cloud CDN
Security Groups	Stateful firewall rules	NSGs (Network Security Groups)	Firewall rules
Global Routing	Route 53, Global Accelerator	Traffic Manager, Front Door	Cloud Load Balancing (native global)

MAIN CLOUD SERVICES COMPARISONS:
STORAGE PHILOSOPHY

Dimension	AWS	Azure	GCP
Object Storage	S3 (industry standard)	Blob Storage	Cloud Storage
Storage Tiers	S3 Standard, IA, Glacier (multiple)	Hot, Cool, Archive	Standard, Nearline, Coldline, Archive
Durability	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)
Consistency	Strong consistency	Strong consistency	Strong consistency
File Storage	EFS, FSx	Azure Files, NetApp Files	Filestore
Block Storage	EBS (many types)	Managed Disks	Persistent Disks, Local SSD
Data Transfer	High egress costs	High egress costs	Transfer Appliance, lower egress for some
Lifecycle Mgmt	Comprehensive	Good	Good

MAIN CLOUD SERVICES COMPARISONS:
DATABASE OFFERINGS

Dimension	AWS	Azure	GCP
Relational	RDS, Aurora	SQL Database, PostgreSQL, MySQL	Cloud SQL, AlloyDB
Standout DB	Aurora (MySQL/PostgreSQL compatible, fast)	Azure SQL (best for SQL Server)	Cloud Spanner (global consistency)
NoSQL Document	DocumentDB (MongoDB compatible)	Cosmos DB (multi-model, excellent)	Firestore
Key-Value	DynamoDB (mature)	Table Storage, Cosmos DB	Cloud Bigtable, Firestore
In-Memory	ElastiCache (Redis, Memcached)	Cache for Redis	Memorystore (Redis, Memcached)
Graph	Neptune	Cosmos DB (Gremlin API)	No native offering
Time-Series	Timestream	Time Series Insights, Cosmos DB	Cloud Bigtable
Global Distribution	Aurora Global Database	Cosmos DB (excellent)	Cloud Spanner (best consistency)
Open-Source	Strong support	Good support	Strong PostgreSQL focus

MAIN CLOUD SERVICES COMPARISONS:
MANAGEMENT AND OPERATIONS

Dimension	AWS	Azure	GCP
Console	AWS Management Console (feature-rich, complex)	Azure Portal (modern, intuitive)	Google Cloud Console (clean, simple)
CLI	AWS CLI (comprehensive)	Azure CLI, PowerShell (excellent)	gcloud CLI (developer-friendly)
APIs	REST APIs, comprehensive	REST APIs, comprehensive	REST APIs, gRPC support
IaC Native	CloudFormation	ARM Templates, Bicep	Cloud Deployment Manager
IaC 3rd Party	Terraform (excellent), CDK, Pulumi	Terraform (excellent), Pulumi	Terraform (excellent), Pulumi
Resource Hierarchy	Organizations > Accounts > Resources	Management Groups > Subscriptions > Resource Groups	Organizations > Folders > Projects
Tagging	Tags (key-value)	Tags and Resource Groups	Labels (key-value)
Resource Naming	ARN (Amazon Resource Name)	Resource ID	Resource Name (URL-like)
Change Tracking	CloudTrail, Config	Activity Log, Resource Graph	Cloud Logging, Asset Inventory

MAIN CLOUD SERVICES COMPARISONS:
MONITORING AND OBSERVABILITY

Dimension	AWS	Azure	GCP
Monitoring	CloudWatch	Azure Monitor	Cloud Monitoring (Operations)
Logging	CloudWatch Logs	Log Analytics	Cloud Logging
Tracing	X-Ray	Application Insights	Cloud Trace
APM	X-Ray, CloudWatch Application Insights	Application Insights (excellent)	Cloud Trace, Profiler
Dashboards	CloudWatch Dashboards	Azure Dashboards, Workbooks	Cloud Monitoring Dashboards
Alerting	CloudWatch Alarms	Azure Monitor Alerts	Cloud Monitoring Alerts
Log Query	CloudWatch Insights	Kusto Query Language (KQL, powerful)	Cloud Logging Query Language
Integration	Good with AWS services	Excellent across Azure	Good integration
Metrics Retention	15 months	93 days default	6 weeks free, longer paid

MAIN CLOUD SERVICES COMPARISONS:
SECURITY AND COMPLIANCE

Dimension	AWS	Azure	GCP
Compliance Certs	100+ (most comprehensive)	90+ (strong enterprise)	70+ (growing)
Encryption Default	Optional (must enable)	Optional (must enable)	Some services encrypt by default
Key Management	KMS, CloudHSM	Key Vault, Managed HSM	Cloud KMS, Cloud HSM
Threat Detection	GuardDuty	Defender for Cloud, Sentinel	Security Command Center
DDoS Protection	AWS Shield (Standard/Advanced)	DDoS Protection (Basic/Standard)	Cloud Armor
Secrets Mgmt	Secrets Manager, Parameter Store	Key Vault	Secret Manager
Vulnerability Scanning	Inspector	Defender for Cloud	Security Command Center
Network Security	Security Groups, NACLs, WAF	NSGs, Application Gateway WAF	VPC Firewall Rules, Cloud Armor
Data Classification	Macie	Purview	Data Loss Prevention API
Security Maturity	Most mature	Very mature, enterprise-focused	Maturing rapidly

MAIN CLOUD SERVICES COMPARISONS:
INTEGRATION AND ECOSYSTEM

Dimension	AWS	Azure	GCP
Marketplace	AWS Marketplace (largest)	Azure Marketplace (comprehensive)	Google Cloud Marketplace (growing)
Partners	Largest partner network	Strong enterprise partners	Growing partner ecosystem
ISV Support	Broadest support	Excellent enterprise ISV support	Good SaaS integration
Open Source	Good support	Good support	Strong open-source DNA
API Gateway	API Gateway, App Sync	API Management	Apigee (industry leading), Cloud Endpoints
Integration Platform	EventBridge, SNS, SQS	Service Bus, Event Grid, Logic Apps	Pub/Sub, Workflows
Cross-Service	Generally good	Excellent (unified approach)	Very good

MAIN CLOUD SERVICES COMPARISONS:
AI/ML AND INNOVATION

Dimension	AWS	Azure	GCP
Marketplace	AWS Marketplace (largest)	Azure Marketplace (comprehensive)	Google Cloud Marketplace (growing)
Partners	Largest partner network	Strong enterprise partners	Growing partner ecosystem
ISV Support	Broadest support	Excellent enterprise ISV support	Good SaaS integration
Open Source	Good support	Good support	Strong open-source DNA
API Gateway	API Gateway, App Sync	API Management	Apigee (industry leading), Cloud Endpoints
Integration Platform	EventBridge, SNS, SQS	Service Bus, Event Grid, Logic Apps	Pub/Sub, Workflows
Cross-Service	Generally good	Excellent (unified approach)	Very good

MAIN CLOUD SERVICES COMPARISONS:
HYBRID AND MULTI CLOUD

Dimension	AWS	Azure	GCP
On-Premise Extension	AWS Outposts	Azure Stack (Hub, HCI, Edge) - most mature	Anthos (strong Kubernetes focus)
Hybrid Strategy	Limited, cloud-first	Best-in-class hybrid (Microsoft legacy)	Container/Kubernetes-centric
Edge Computing	Wavelength, Local Zones, Outposts	Azure Edge Zones, Stack Edge	Distributed Cloud, Edge locations
Multi-Cloud Tools	Limited native support	Arc (manage resources across clouds)	Anthos (run on AWS, Azure, on-prem)
Container Portability	ECS (proprietary), EKS (Kubernetes)	AKS (Kubernetes)	GKE (best Kubernetes), Anthos
Migration Tools	Application Migration Service, DMS	Azure Migrate (comprehensive)	Migrate for Compute Engine, DMS
Hybrid Networking	Direct Connect, VPN	ExpressRoute (mature)	Cloud Interconnect

MAIN CLOUD SERVICES COMPARISONS:
SUPPORT AND SLA

Dimension	AWS	Azure	GCP
Support Tiers	Developer, Business, Enterprise	Developer, Standard, Professional Direct, Premier	Basic, Standard, Enhanced, Premium
Response Times	15 min (critical, Enterprise)	15 min (critical, Premier)	15 min (critical, Premium)
Pricing	3-10% of monthly spend	3-10% of monthly spend	3-10% of monthly spend
TAM	Enterprise Support	Premier Support	Premium Support
Service SLAs	99.9-99.99% typical	99.9-99.99% typical	99.9-99.99% typical
Uptime History	Generally excellent	Generally excellent	Occasional high-profile outages
Community	Largest forums, re:Post	Active community, Q&A	Stack Overflow, Google Groups
Documentation Quality	Comprehensive, sometimes outdated	Good, well-maintained	Generally excellent, clear

MAIN CLOUD SERVICES COMPARISONS:
VENDOR LOCK-IN RISK

Dimension	AWS	Azure	GCP
Proprietary Services	Many (DynamoDB, Lambda specifics)	Many (Cosmos DB, Azure Functions)	Fewer (more open-source aligned)
Open Standards	Increasing support	Good support	Strong commitment
Container Standards	ECS proprietary, EKS standard	AKS standard	GKE standard, strong Kubernetes
Data Portability	Good but egress costs high	Good but egress costs high	Good, slightly better egress
API Lock-in	Service-specific APIs	Service-specific APIs	More standardized APIs
Exit Tools	Limited native tools	Azure Migrate can help	Transfer Appliance
Multi-Cloud Strategy	Not encouraged	Arc enables multi-cloud	Anthos designed for multi-cloud

MAIN CLOUD SERVICES COMPARISONS:
MATURITY

Dimension	AWS	Azure	GCP
Market Share	~32% (leader)	~23% (strong #2)	~10% (solid #3)
Launch Year	2006 (pioneer)	2010 (fast follower)	2008 (late to market)
Service Count	200+ services	200+ services	100+ services
Innovation Pace	Very fast, thousands of features/year	Fast, catching up	Fast, focused innovation
Service Maturity	Most mature overall	Rapidly maturing	Newer but stable
Financial Backing	Amazon	Microsoft	Google/Alphabet
Growth Rate	Steady, slowing	Fastest growing	Growing steadily
Customer Types	Startups to enterprise	Strong enterprise focus	Tech companies, growing enterprise
Enterprise Adoption	Highest	Rapidly growing	Growing

MAIN CLOUD SERVICES COMPARISONS:
SPECIALIZED CAPABILITIES

Dimension	AWS	Azure	GCP
Industry Solutions	Healthcare (HealthLake), Financial Services	Strong in all (especially enterprise verticals)	Healthcare, Retail, Media focus
IoT Platform	IoT Core, Greengrass (comprehensive)	IoT Hub, IoT Central (strong)	IoT Core (simpler)
Gaming	GameLift, Lumberyard	PlayFab (strong)	Game Servers (limited)
Media/Streaming	Elemental Media Services (best)	Media Services	Media CDN, Transcoder
Blockchain	Managed Blockchain (Ethereum, Hyperledger)	Blockchain Service (limited)	Blockchain Node Engine
Quantum	Braket	Azure Quantum	Quantum Engine (partner access)
Robotics	RoboMaker	Limited offerings	Cloud Robotics (limited)
Satellite	Ground Station	Azure Orbital	No native offering
HPC	ParallelCluster, batch computing	HPC Cache, CycleCloud	HPC Toolkit

MAIN CLOUD SERVICES COMPARISONS:
PERFORMANCE AND SCALABILITY

Dimension	AWS	Azure	GCP
Compute Performance	Excellent, many instance types	Excellent, many options	Excellent, custom machines
Network Throughput	Up to 100 Gbps	Up to 200 Gbps (Accelerated Networking)	Up to 100 Gbps
Storage IOPS	Up to 256,000 (io2 Block Express)	Up to 160,000 (Ultra Disk)	Up to 100,000 (Extreme PD)
Object Storage Speed	S3 Transfer Acceleration, multi-part upload	Hot tier high performance	Global load balancing, CDN integration
Scaling Limits	Very high (varies by service)	Very high (varies by service)	Very high (varies by service)
Cold Start Times	Lambda: 100ms-1s	Functions: 100ms-1s	Cloud Functions: 100ms-1s, Cloud Run: faster
Database Performance	Aurora up to 5x MySQL performance	Azure SQL Hyperscale (high performance)	Cloud Spanner (consistent global performance)
Auto-scaling Speed	Fast (seconds to minutes)	Fast (seconds to minutes)	Fast (seconds to minutes)
Global Latency	Good, many PoPs	Good, extensive network	Excellent (Google's backbone network)

MAIN CLOUD SERVICES COMPARISONS:
EXIT STRATEGY

Dimension	AWS	Azure	GCP
Data Export	S3 APIs, Data Pipeline, DataSync	Storage Explorer, AzCopy, Data Factory	gsutil, Storage Transfer Service
Migration Out Tools	Limited, third-party tools needed	Azure Migrate (can work both ways)	Transfer Appliance, third-party tools
Egress Costs	High (up to $0.09/GB)	High (up to $0.087/GB)	High (up to $0.08/GB), some free tiers
Data Transfer Appliances	Snowball, Snowmobile	Data Box	Transfer Appliance
Contract Terms	Pay-as-you-go default, no lock-in	Pay-as-you-go default, Enterprise Agreements	Pay-as-you-go default, no lock-in
Service Termination	Keep running until you delete	Keep running until you delete	Keep running until you delete
Data Retention Post-Cancel	Varies by service (typically 30-90 days)	Varies by service (typically 90 days)	Varies by service (typically 30 days)
Backup to Another Cloud	Supported via third-party tools	Supported via third-party tools	Supported via third-party tools
Documentation for Exit	Limited official guidance	Some guidance available	Limited official guidance

MAIN CLOUD SERVICES COMPARISONS: SUMMARY & KEY DIFFERENTIATORS

AZURE STRENGTHS:

🏢 Best for enterprises (especially with Microsoft stack)

🔄 Best hybrid cloud capabilities (Azure Stack, Arc)

🤝 Seamless integration with Office 365, Active Directory

🎯 Azure OpenAI exclusive partnership

💻 Superior Windows workload support

AWS STRENGTHS:

🏆 Market leader with most mature services

🔧 Broadest service portfolio (200+ services)

🌍 Largest partner ecosystem

💼 Best for startups and general-purpose cloud

📊 Most comprehensive offerings across all categories

GCP STRENGTHS:

🧠 Best for data analytics and ML/AI innovation

🌐 Superior global network (Google's backbone)

☸️ Best Kubernetes experience (GKE, Anthos)

💰 More transparent pricing, sustained use discounts

🔓 Strong open-source and multi-cloud commitment

📈 BigQuery unmatched for analytics

MAIN CLOUD SERVICES COMPARISONS: WHEN TO CHOOSE EACH?

Choose Azure if:

•You're already in the Microsoft ecosystem
•You need strong hybrid cloud capabilities
•You're an enterprise with on-premise infrastructure
•You need the best enterprise support and compliance

Choose GCP if:

•You're doing heavy data analytics or ML/AI
•You want the best Kubernetes and container experience
•You need global low-latency applications
•You value open standards and multi-cloud flexibility
•You're building data-intensive applications

Choose AWS if:

•You need the broadest service selection
•You want the most mature ecosystem
•You're building a startup or general SaaS
•You need specialized services (IoT, media, satellite)

COSTS ENGINEERING

SUBJECT CURRICULUM

1 - Data Engineering Fundamentals

- 1.1 - What is Data Engineering
- 1.2 - Data Lifecycle
- 1.3 - Programming Languages
  - 1.3.1 - Python, R y Jupyter Notebooks
  - 1.3.2 - Other Languages for Data Engineering
- 1.4 - EL vs ETL vs ELT
- 1.5 - Data Governance

2 - Cloud Services

- 2.1 - Cloud vs On Premise
- 2.2 - Cloud Services Comparison
- 2.3 - Costs Engineering
- 2.4 - Specific Cloud Providers

3 - Distributed Systems

- 3.1 - Distributed Storage Systems
- 3.2 - Distributed Computing
- 3.3 - Event-Based architectures
- 3.4 - Orchestration
- 3.5 - Serverless Architectures

EXERCISE

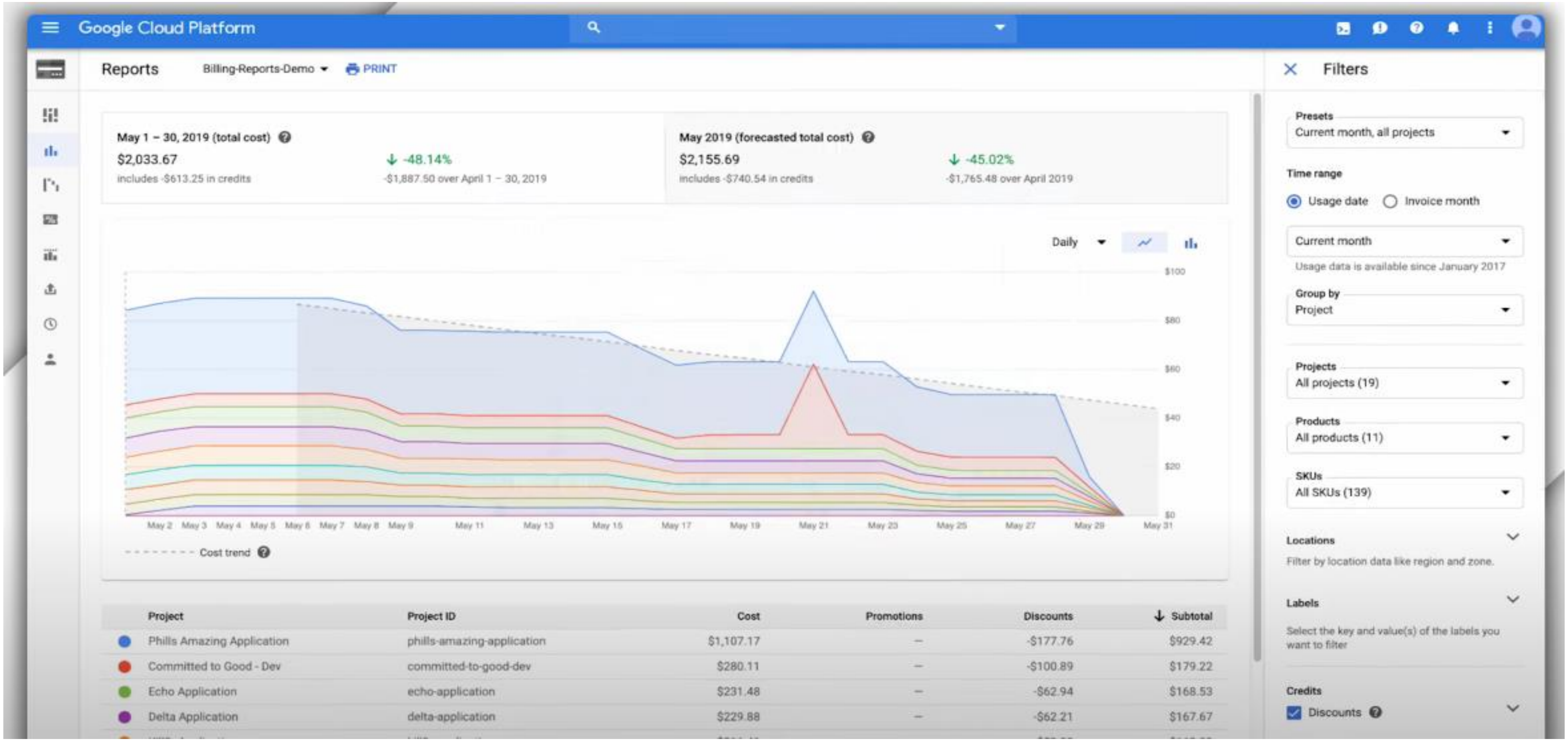
Select one of the cloud providers.

Think about this scenario: a daily pipeline that gets 24GB of CSV data from an API (So we can use python to download it). We need to make a validation process before ingesting it in a Database for analytical purposes and consumed by about 100 people daily.

Design (draw a diagram) of the solution.

Use the cost calculator of the selected platform to estimate the monthly cost of this solution.

COST MONITORING



OTHER CLOUD PROVIDERS

SUBJECT CURRICULUM